

CHAPTER 6 Hypothesis Testing

In hypothesis testing, we test an **assumed** (hypothesised) value of a population parameter. Using some sample information, we may want to know whether **a given claim** (or statement) about a population parameter **is true or not**. Then, we can either **accept or reject** the hypothesised value.

Statistical Hypothesis

- ⇒ A statement, assumption or belief about a population parameter(s)
- ⇒ Experimental or sample evidence is required to verify the statement

Null Hypothesis (H_0)

- ⇒ A claim (or statement) about a population parameter that is **assumed to be true** until it is declared false
- ⇒ Specifies the value of the population parameter to be tested

Alternative Hypothesis (H_1)

- ⇒ A claim (or statement) about a population parameter that **will be true if the null hypothesis is false**
- ⇒ The **rejection of H_0** means to **accept** the alternative hypothesis H_1

Results of hypothesis testing

There are only four possible results when we test a given hypothesis.

1. We accept a true hypothesis
 - ⇒ a correct decision
2. We reject a false hypothesis
 - ⇒ a correct decision
3. We **reject a true hypothesis**
 - ⇒ an incorrect decision
 - ⇒ known **as Type I error** (denoted by α --- “alpha”)
4. We **accept a false hypothesis**
 - ⇒ an incorrect decision
 - ⇒ known **as Type II error** (denoted by β --- “beta”)

Significance Levels

- ⇒ The **maximum probability** of **making Type I error** in hypothesis testing

- ⇒ Denoted by α
- ⇒ Usually specified before a hypothesis test is made
- ⇒ The value of 5% ($\alpha = 0.05$) or 1% ($\alpha = 0.01$) is frequently used

Example:

If we select 5% significance level, we will expect that the probability of making an error of rejecting the hypothesis when it is true is 5%. In other words, we are about 95% confidence that we will make a correct decision although we could be wrong with a probability of 5%.

Type I error (denoted by α)

- ⇒ Occurs when a null hypothesis is rejected as false when in fact it is true
- ⇒ $\alpha = P(H_0 \text{ is rejected} \mid H_0 \text{ is true})$
- ⇒ The value of α represents the significance level of the test

Type II error (denoted by β)

- ⇒ Occurs when a null hypothesis is accepted as true when in fact it is false
- ⇒ $\beta = P(H_0 \text{ is not rejected} \mid H_0 \text{ is false})$
- ⇒ The value of β represents the probability of committing a Type II error
- ⇒ The value of $1 - \beta$ is called the power of the test; it represents the probability of not making a Type II error

Critical Region / Rejection Region

- ⇒ The region which corresponds to a predetermined levels of significance
- ⇒ When the sample statistic falls in the critical region, we reject the hypothesis as it will be considered to be false

The critical region may be represented by a portion of the area under the normal curve in two ways:

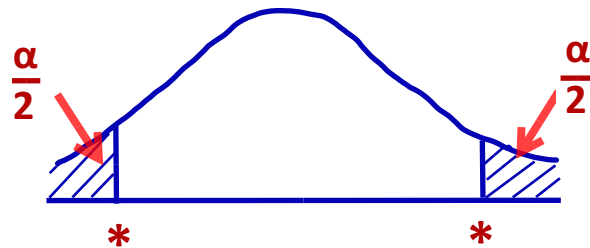
1. Two tails or sides under the curve
2. One tail or side under the curve which is either the right tail or left tail

Two-tailed test

- ⇒ The test of hypothesis which are based on a critical region represented by both tails under the normal curve

Two-tailed test

--> The test of hypothesis which are based on a critical region represented by both tails under the normal curve.

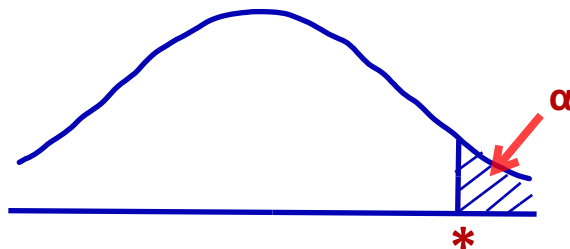


 Rejection Region  Acceptance Region * Critical Value

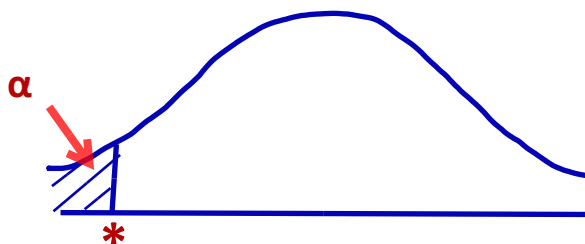
One-tailed test

⇒ The test of hypothesis which are based on a critical region represented by only one tails under the normal curve

★ Right-tailed test



★ Left-tailed test



 Rejection Region  Acceptance Region * Critical Value

Critical value

⇒ The value that separates the rejection region from the acceptance region

Sign of H_1	Type of test
$>$	Right-tailed test
$\neq$	Two-tailed test
$<$	Left-tailed test

There are different notations for mean, standard deviation, proportion and size of a population and a sample.

Statistic	Population	Sample
Mean	μ	$\bar{X}$
Standard deviation	σ	S
Proportion	π	P_s
Size	N	n

Steps to perform a hypothesis testing

1. State the null and alternative hypothesis
2. Determine the significance level α and the critical value
3. Select the distribution (test statistic) to use
4. Determine the rejection and non-rejection regions
5. Calculate the value of the test statistic
6. Make a decision (reject H_0 or fail to reject H_0)

Test Statistic

⇒ can be defined as a rule/criterion to determine acceptance/rejection of H_0

Hypothesis testing about a population mean: Large sample

★ The null and alternative hypothesis

	Null Hypothesis H_0	Alternative Hypothesis H_1	Type of test
(i)	$\mu = \mu_0$	$\mu \neq \mu_0$	Two-tailed test
(ii)	$\mu = \mu_0$ or $\mu \geq \mu_0$	$\mu < \mu_0$	Left-tailed test
(iii)	$\mu = \mu_0$ or $\mu \leq \mu_0$	$\mu > \mu_0$	Right-tailed test

★ Test statistic

i) $z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$ if σ is known

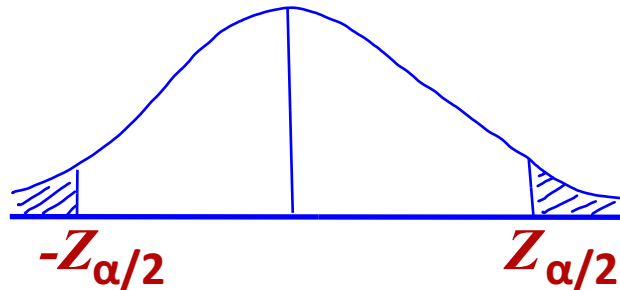
ii) $z = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$ if σ is unknown and $n \geq 30$

★ Critical value and rejection region

	H_0	H_1 ✓	Critical Value	Critical Region ^{rejection}
(i)	$\mu = \mu_0$	$\mu \neq \mu_0$	$\pm Z_{\alpha/2}$	$Z < -Z_{\alpha/2}$ or $Z > Z_{\alpha/2}$
(ii)	$\mu \geq \mu_0$	$\mu < \mu_0$	$-Z_\alpha$	$Z < -Z_\alpha$
(iii)	$\mu \leq \mu_0$	$\mu > \mu_0$	Z_α	$Z > Z_\alpha$

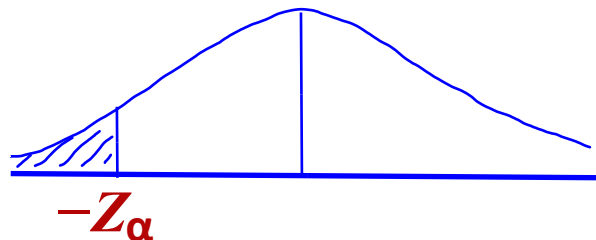
(i) Two-tailed test

when " $\neq$ "



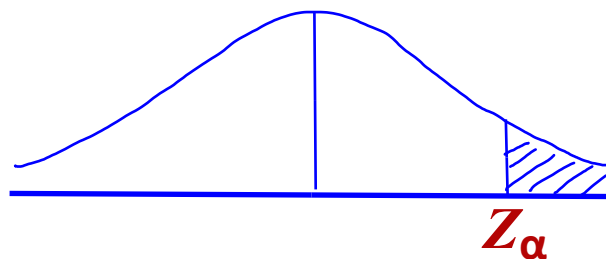
(ii) Left-tailed test

when "<"



(iii) Right-tailed test

when ">"



Rejection Region

- reject H_0 / Accept H_1



Non-rejection / Acceptance Region

- do not reject H_0

Example 1:

A company markets car tyres. Their lives are normally distributed with a mean of 40,000 km and standard deviation of 3,000 km. A change in the production process is believed to result in a better product. A test sample of 64 new tyres has a mean life of 41,200 km. Can you conclude that the new product is significantly better than the current one? ($\alpha = 0.05$) ^{claim}

Solution: $\mu = 40000$ $\sigma = 3000$ $n = 64$ $\bar{x} = 41200$

Let μ = true mean life of new tyres

$H_0: \mu \leq 40000$

$H_1: \mu > 40000$ claim

At $\alpha = 0.05$ critical value: $Z_{\alpha} = Z_{0.05} = 1.6449$
 rejection region: $Z > 1.6449$

Given $\mu_0 = 40000$, $\sigma = 3000$, $n = 64$, $\bar{X} = 41200$

Let $\mu \equiv$ true mean life of the new tires.

$\times H_0: \mu \leq 40000$ (mean life of the new tires are not better than the old tires)

claim $\checkmark H_1: \mu > 40000$ (mean life of the new tires are better than the old tires) (claim)

At $\alpha = 0.05$, critical value $Z_{\alpha} = Z_{0.05} = 1.6449$

Rejection region $Z > 1.6449$



$$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{41200 - 40000}{\frac{3000}{\sqrt{64}}} = 3.2$$

Since $Z = 3.2 > 1.6449$, the computed value Z fall within rejection region, we reject H_0 and accept H_1 and conclude that the new tires are better than the old tires at level $\alpha = 0.05$.

the claim is true

Example 2:

A large retailer wants to determine whether the mean income of families living within two miles of a proposed building site exceeds RM14400. What can he conclude at the 0.05 level of significance, if the mean income of a random sample of 60 families living within two miles of the proposed site is RM14524 and the standard deviation is RM763?

Solution: $\mu = 14400$ $n = 60$, $s = 763$ $\bar{x} = 14524$
 let $\mu \equiv$ true mean income families living within two miles.

$H_0: \mu \leq 14400$

$H_1: \mu > 14400$ (claim)

At $\alpha = 0.05$, critical value, $Z_{\alpha} = Z_{0.05} = 1.6449$
 rejection region, $Z > 1.6449$

Test statistic

$$Z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{14524 - 14400}{\frac{763}{\sqrt{60}}} = 1.2588$$

Since $1.2588 < 1.6449$, failed to reject H_0 at $\alpha = 0.05$ and concluded that the claim is false.

Since $Z = 1.259 < 1.6449$, H_0 cannot be rejected. Hence there is no evidence to suggest that the true mean income of families living within two miles of a proposed site exceeds RM14400 at level $\alpha = 0.05$.

Example 3:

It is thought that a certain Normal population has a mean of 1.6. A sample of 50 gives a mean of 1.51 and a standard deviation of 0.3. Does this provide evidence, at the 5% level, that the population mean is less than 1.6? *claim*

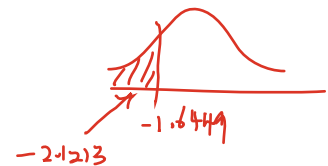
Solution: $\mu = 1.6, S = 0.3, n = 50, \bar{X} = 1.51$

Let μ = true mean for certain normal population

$$H_0: \mu \geq 1.6$$

$$H_1: \mu < 1.6 \text{ (claim)}$$

At $\alpha = 0.05$ critical value, $-Z_{\alpha} = -Z_{0.05} = -1.6449$
rejection region, $Z < -1.6449$



Test statistic

$$Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{1.51 - 1.6}{\frac{0.3}{\sqrt{50}}} = -2.1213$$

Since $Z = -2.1213 < -1.6449$, we reject H_0 at level $\alpha = 0.05$ and conclude that (the population mean is less than 1.6.) the claim is true

Example 4:

The expected lifetime of electric light bulbs produced by a given process was 1500 hours. To test a new batch, a sample of 40 was taken which showed a mean lifetime of 1410 hours. The standard deviation is 90 hours. Test the hypothesis that the mean lifetime of the electric light bulbs has not changed, using a level of significance of 0.05. α

Solution: $\mu = 1500, n = 40, \bar{X} = 1410, S = 90$

Let μ = True mean of electric light bulbs.

$$H_0: \mu = 1500 \text{ (claim)}$$

$$H_1: \mu \neq 1500$$

At $\alpha = 0.05$ critical value, $\pm Z_{\frac{\alpha}{2}} = \pm Z_{0.025} = \pm 1.96$
rejection region, $Z < -1.96$ or $Z > 1.96$



Test statistic

$$Z = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}} = \frac{1410 - 1500}{\frac{90}{\sqrt{40}}} = -6.3246$$

Since $Z = -6.3246 < -1.96$, we reject H_0 at level $\alpha = 0.05$ and conclude that (there is evidence to suggest that the mean life time of the electric light bulbs has changed.) the claim is false.

Example 5:

The management of a club **claims that** its members lose an average of **2 kilograms or more** within the first month after joining the club. A consumer agency wanted to check this claim took a random sample of 36 members of this health club and found that they lost an average of 1.8 kilograms within the first month of membership with a standard deviation of 0.5 kilograms. **Test** the management's **claim** at the 0.05 level of significance.

Solution:

$$\begin{aligned} \mu &= 2 \quad n = 36 \quad \bar{x} = 1.8 \quad s = 0.5 \\ H_0 &: \mu \geq 2 \text{ (claim) } \times \\ H_1 &: \mu < 2 \\ \text{At } \alpha &= 0.05 \quad \text{critical value: } -Z_{\alpha} = -Z_{0.05} = -1.6449 \\ \text{rejection region: } Z &< -1.6449 \\ \text{Test statistic} \\ Z &= \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{1.8 - 2}{\frac{0.5}{\sqrt{36}}} = -2.4 \end{aligned}$$

Since $Z = -2.4 < -1.6449$, we reject H_0 at level $\alpha = 0.05$ and conclude that the management's claim is false not true.

Hypothesis testing about a population proportion: Large sample

★ The null and alternative hypothesis

	Null Hypothesis H_0	Alternative Hypothesis H_1	Type of test
(i)	$\pi = \pi_0$	$\pi \neq \pi_0$	Two-tailed test
(ii)	$\pi \neq \pi_0$ or $\pi \geq \pi_0$	$\pi < \pi_0$	Left-tailed test
(iii)	$\pi \neq \pi_0$ or $\pi \leq \pi_0$	$\pi > \pi_0$	Right-tailed test

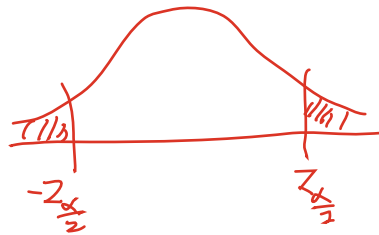
★ Test statistic

$$Z = \frac{P_s - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}} \quad \text{or} \quad Z = \frac{nP_s - n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}} \quad \text{or} \quad Z = \frac{X - n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}}$$

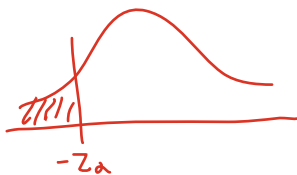
★ Critical value and rejection region

	H_0	H_1	Critical Value	Critical Region
(i)	$\pi = \pi_0$	$\pi \neq \pi_0$	$\pm Z_{\alpha/2}$	$Z < -Z_{\alpha/2}$ or $Z > Z_{\alpha/2}$
(ii)	$\pi \geq \pi_0$	$\pi < \pi_0$	$-Z_{\alpha}$	$Z < -Z_{\alpha}$
(iii)	$\pi \leq \pi_0$	$\pi > \pi_0$	Z_{α}	$Z > Z_{\alpha}$

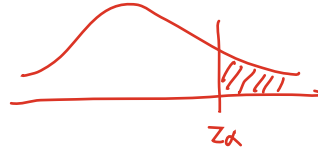
(i) Two-tailed test



(ii) Left-tailed test



(iii) Right-tailed test



Rejection Region

- reject H_0 / Accept H_1



Non-rejection Region

- do not reject H_0

failed to

Note:

1. $\pi_0 \equiv$ the population proportion (predetermined constant)
2. $P_s \equiv$ the sample proportion
3. Population proportion π , is used instead of sample proportion because the population proportion is known

Example 6:

In an investigation into ownership of calculators, 200 randomly chosen school students were interviewed, 163 of them owned a calculator. Using the evidence of this sample, **test** at the 5% level of significance, hypothesis that the proportion of school students owning a calculator is **more than 80%**.

Solution:

Given $n = 200$, $x = 163$, $\pi_0 = 0.8$,
 $P_s = x/n = 163/200 = 0.815$

Let $\pi \equiv$ true proportion of school students
 owning a calculator

$H_0 : \pi \leq 0.8$ (**not more than 80%**)

$H_1 : \pi > 0.8$ (**more than 80%**)

At $\alpha = 0.05$, critical value $Z_{\alpha} = Z_{0.05} = 1.6449$
 Rejection region $Z > 1.6449$

$$Z = \frac{P_s - \pi_0}{\sqrt{\frac{\pi_0(1 - \pi_0)}{n}}} = \frac{0.815 - 0.8}{\sqrt{\frac{0.8(0.2)}{200}}} = 0.53$$

Since $Z = 0.53 < 1.6449$, H_0 cannot
 be rejected and there is no evidence
 to suggest that proportion of school
 students owning a calculator is
 more than 80% at level $\alpha = 0.05$.

Example 7:

An election candidate $\overset{n}{\text{claims that 60 percent}}$ of the voters support him. A random sample of 2500 voters show that 1400 support him. $\overset{0.6}{\text{Test his claim}}$ at 0.10 level of significance. $\times$

Solution: $\pi = 0.6$ $p_s = \frac{x}{n} = \frac{1400}{2500} = 0.56$ $n = 2500$

$$H_0: \pi = 0.6 \text{ (claim)}$$

$$H_1: \pi \neq 0.6$$

At $\alpha = 0.1$ critical value: $\pm Z_{\frac{\alpha}{2}} = \pm Z_{0.05} = \pm 1.6449$
rejection region: $Z < -1.6449$ or $Z > 1.6449$

Test statistic

$$Z = \frac{p_s - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.56 - 0.6}{\sqrt{\frac{0.6(0.4)}{2500}}} = -4.08$$

Since $Z = -4.08 < -1.6449$, we
✓ reject H_0 at level $\alpha = 0.1$ and
conclude that the election
candidate's claim is not true.

false

Example 8:

A coin is tossed 100 times and 38 heads are obtained. Is there evidence, at the 2% level that the coin is biased in favour of tails?

Solution: $\pi = 0.5$ $p_s = \frac{x}{n} = \frac{38}{100} = 0.38$ $n = 100$

$$H_0: \pi \geq 0.5$$

$$H_1: \pi < 0.5 \text{ (claim)}$$

At $\alpha = 0.02$ critical value: $-Z_{\alpha} = -Z_{0.02} = -2.0537$
rejection region: $Z < -2.0537$

Test statistic

$$Z = \frac{p_s - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.38 - 0.5}{\sqrt{\frac{0.5(0.5)}{100}}} = -2.4$$

Since $Z = -2.4 < -2.0537$, reject H_0 at $\alpha = 0.02$
and conclude that the claim is true.

Example 9:

Solution:

Example 9:

ABC Mailing Company sells computers and computer parts by mail. The company claims that at least 90% of all orders are mailed within 72 hours after they are received. The quality control department at the company often takes samples to check if this claim is valid. A recently taken sample of 150 orders showed that 129 of them were mailed within 72 hours. Do you think the company's **claim is true**? Use a 2.5% significance level.

$$\pi = 0.9 \quad n = 150 \quad P_s = \frac{X}{n} = \frac{129}{150} = 0.86$$

$$H_0: \pi \geq 0.9 \quad (\text{claim})$$

$$H_1: \pi < 0.9$$

$$\text{At } \alpha = 0.025$$

$$\text{critical value} = -Z_{\alpha} = -Z_{0.025} = -1.96$$

$$\text{rejection region} : Z < -1.96$$

Test statistic

$$Z = \frac{P_s - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} = \frac{0.86 - 0.9}{\sqrt{\frac{0.9(1-0.9)}{150}}} = -1.63$$

Since $Z = -1.63 > -1.96$, failed to reject H_0 at $\alpha = 0.025$ and conclude that the claim is true